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Dr. M. S. Sheshgiri Campus, Belagavi

Digital Image Processing

"Brain Tumor Detection Using Image Processing"

ELECTRONICS AND COMMUNICATION ENGINEERING

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PROBLEM STATEMENT

Brain tumor detection is crucial for early diagnosis and treatment, but manual MRI analysis is time-consuming. This project aims to develop an automated detection system using digital image processing techniques like segmentation and feature extraction. The solution enhances accuracy, speeds up diagnosis, and assists radiologists in identifying tumors more efficiently.

INTRODUCTION

Brain tumors are a critical medical concern requiring early and accurate detection to improve patient outcomes. Magnetic Resonance Imaging (MRI) is the gold standard for brain imaging, but manual tumor detection and segmentation can be time-consuming and subjective. This project presents an automated pipeline that processes MRI images to extract the brain region and detect potential tumor regions using classical image processing techniques, including adaptive thresholding, morphological operations, and statistical analysis. The pipeline also calculates a confidence score for tumor detection and provides clear visual outputs to support clinical decision-making.

OBJECTIVES

1. Brain Segmentation: Isolate the brain region from the MRI scan using image processing techniques.

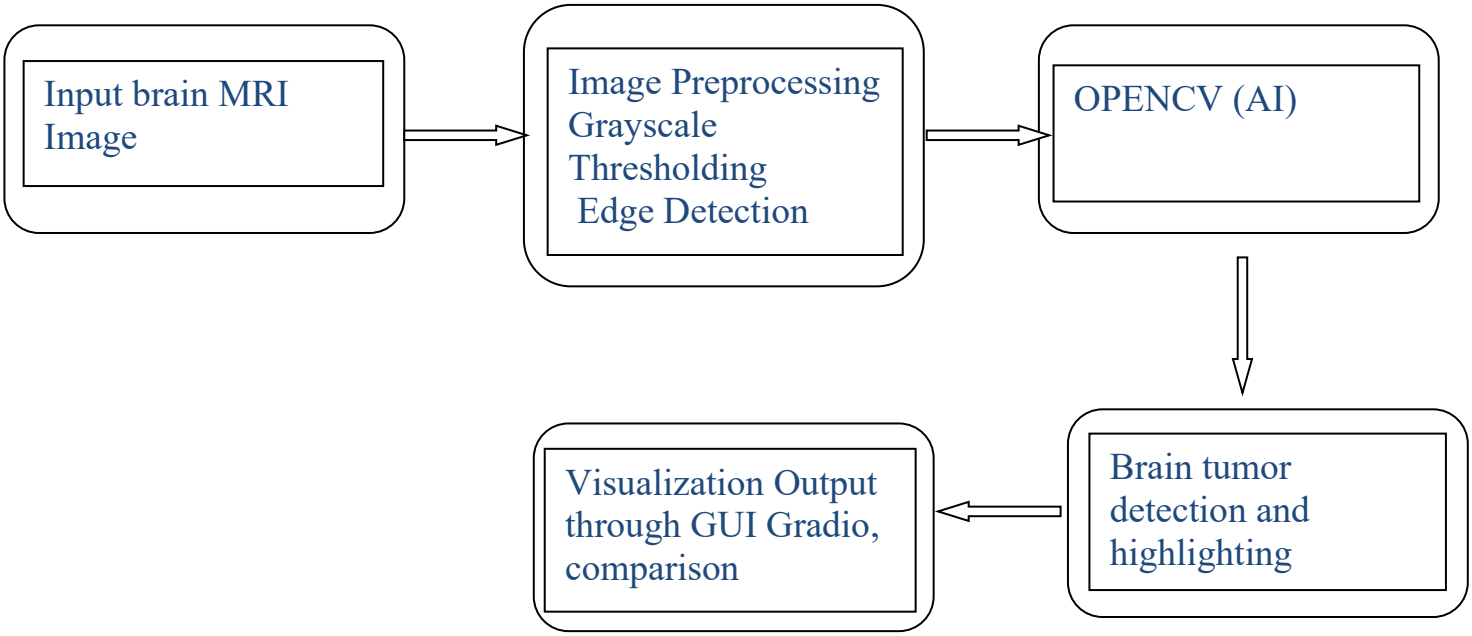
2. Tumor Detection: Identify potential tumor areas using intensity, texture, and statistical analysis.

3. Confidence Scoring: Calculate how confident the system is about detecting a tumor.

4. Visual Outputs: Generate clear images showing brain extraction, tumor detection, and overlays.

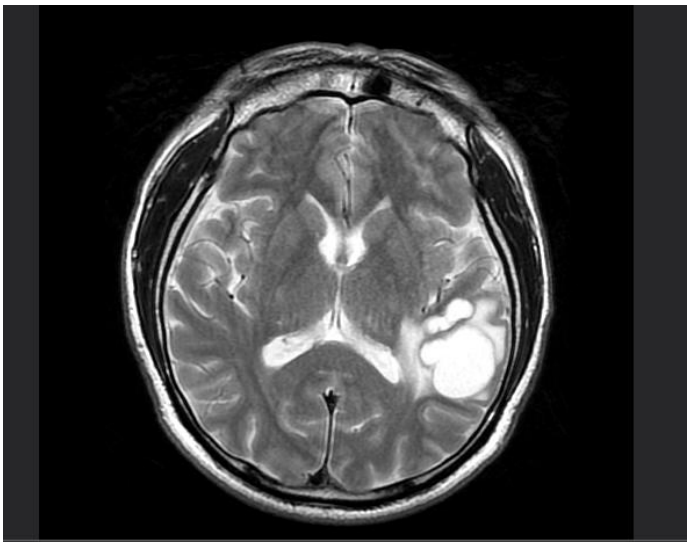
5. User Interface: Provide a user-friendly interface using Gradio for easy image upload and result visualization.

i. BLOCK DIAGRAM

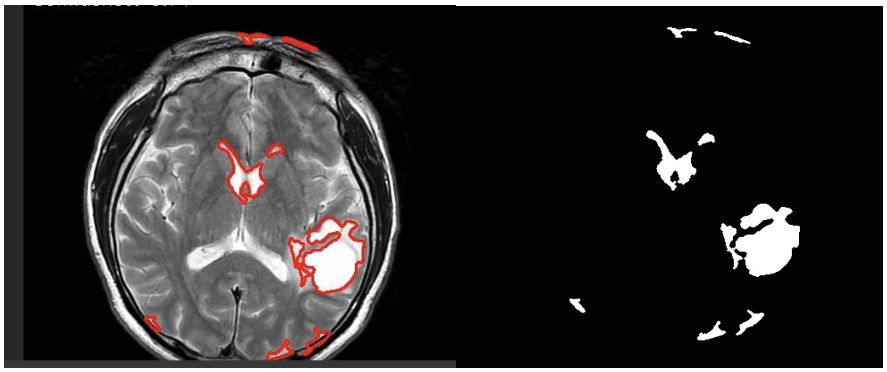


RESULTS

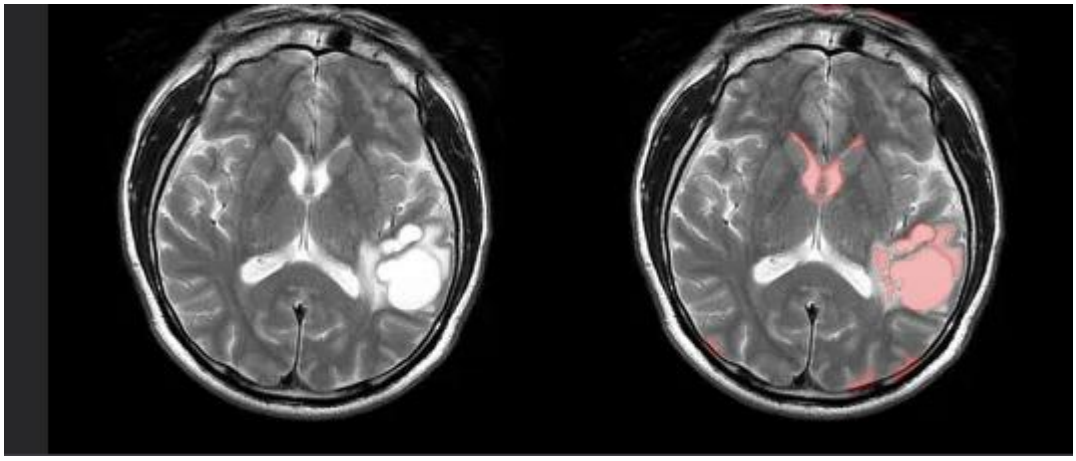
1. INPUT IMAGE



2. PROCESSED IMAGE



2. **OUTPUT IMAGE**



CONCLUSION

This pipeline demonstrates a classic computer vision approach for brain tumor detection using MRI images. It leverages adaptive thresholding, morphological processing, and statistical techniques to segment the brain and detect potential tumor regions. While it does not use a machine learning model, it provides a quick and interpretable way to identify suspicious areas and assess detection confidence. This approach is efficient, easy to deploy, and suitable for preliminary analysis or as a baseline system for further development with more advanced deep learning methods